

traffic infrastructure automatically based on different situations.

These capabilities can be beneficial in various traffic operations, starting with real-time surveillance of traffic rules. Through computer vision and object recognition, AI platforms can analyse live video feeds collected through surveillance systems and sensors to auto-detect a vehicle that may break a road rule. The platform can detect vehicle type and registered vehicle number so that necessary actions can be taken.

Administratives at traffic control stations not only receive live traffic status of individual areas, the system also provides them with predictive traffic estimations, suggestions of alternate routes and comparative insights of traffic conditions across parameters. These can be utilised to direct traffic in a planned manner.

Deep learning algorithms enable AI platforms to automatically control traffic signals—main data being fed through computer vision, satellite data and analytics. There are other benefits that AI can bring to the roads. Autonomous cars, which will soon be a part of the connected ecosystem, can be ensured to be safe on the road with AI-powered traffic control systems. Emergency responses can be greatly improved through automated alerts.

Deployment stories

There are many interesting deployment cases around the globe. For instance, towards the start of this year, streets in Kuala Lumpur, Malaysia, deployed a traffic management solution called City Brain, created by Alibaba group. The platform was deployed utilising 300 traffic lights, 500 CCTV cameras and other public transport infrastructure. It is enabling automated traffic control, road surveillance and optimal route detection.

The platform also draws in live feed from social media and other sources. The pilot was deployed in partnership with Malaysia Digital Economy Corp. and Kuala Lumpur City Hall.

Before this, City Brain was launched in Hangzhou, China in 2016. Since then, it has expanded to 1300 traffic lights and 420-sqkm stretches of coverage area. Reports suggest a 15 per cent increase in traffic speed, reduced congestion and

traffic rule violation detection with at least 92 per cent accuracy.

In September 2018, Hangzhou government deployed City Brain 2.0, which will also assist in city-level management, like real-time response to fire alarms, road planning and medical emergencies. It is estimated that the arrival time of fire trucks can be improved by 49 per cent with this platform.

Madison, Wisconsin, USA, has been using an intelligent traffic management platform since 2014, to mitigate its expected increase in traffic volume over the coming years and to reduce the increasing number of road accidents. Wisconsin Department of Transportation and City of Madison have deployed a traffic management platform called Centracs Adaptive, by Econolite.

Vehicle detection sensors like inductive loops, above-ground video and microwave radar detection sensors and video detection systems have been installed to transmit traffic data to Centracs Adaptive module. The result is at least 22 per cent reduction in travel time, 65 per cent reduction in vehicle stops and substantially reduced accidents at intersections. Alternate routes for closed roads and controlling traffic spikes has also become easier.

Pilots in India

Indian cities are finally up in arms about tackling the massively challenging traffic situations. For instance, Electronic City Township Authority (ELCITA), Bengaluru, has partnered with Siemens Corporate Technology to deliver intelligence to its traffic control system around the Electronic City area. Trials are currently on, which are expected to deliver automated traffic light control, traffic density calculation, vehicle detection, auto-ticket generation, incident detection, quick response, green corridor arrangement and so on. Siemens team has partnered with IISC, Bengaluru, to further scale this platform.

Delhi Traffic Police is also gearing up for an AI-driven traffic management system by April 2019, after receiving approval from Ministry of Home Affairs (MHA). With a tentative investment of ₹ 10 billion, the police will utilise smart signals, intelligent

CCTV cameras, automated ticketing systems, AI software and cloud analytics, among others, to gain deeper and consistent traffic data.

Kolkata Police has proposed and tested an algorithm using Google Live Traffic data to solve the challenge of disorganised, standalone traffic signal operation and congestion. For the same, signals have been inter-networked using Wi-Fi. Dynamic control of connected signals aims to deliver organised signal timings.

Based on satellite data, CCTV data and data generated by Google, the platform is set to monitor traffic volume and plan traffic motion accordingly. If a traffic stretch extends beyond 150 metres, nearest officers in charge will be notified to handle the situation.

Traffic management has collateral benefits of reduced environmental pollution, improved mobility and transport systems, and an overall improved way of life for the citizens. With the right technology and talent in place, an intelligent traffic management should not be far away.

—Paromik Chakraborty, technical journalist, EFY

“Over 90 Per Cent Of SME Physical Retail Stores Do Not Connect With Consumers”

➤ Today, retail businesses need a combination of physical stores and digital platforms to deliver shoppers the optimal experience. In this context, DLF Group recently deployed Phygital platform across five of its malls.